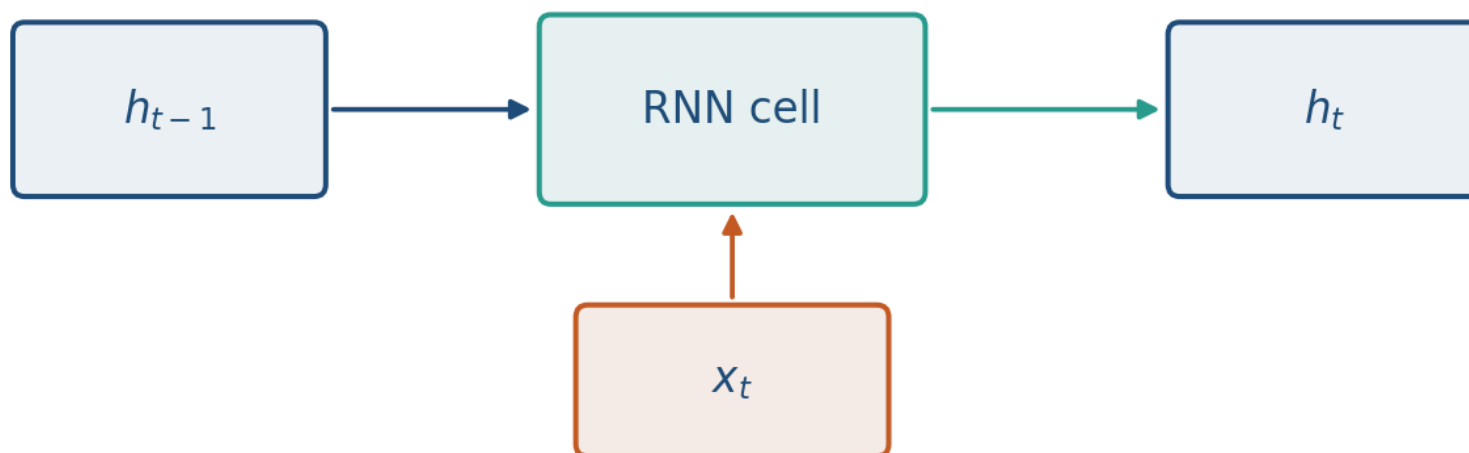


A cell with memory

- Each step: x_t and h_{t-1} become h_t .
- The same weights at every token.
- Reset the state between unrelated documents.

Same weights at every step: input plus previous state



Unrolled in time

- The loop is a chain. That chain is the long-range path.
- A CNN is parallel with a fixed window. An RNN is sequential with the whole past.

Unrolled in time. The chain is why long-range gradients struggle.

